higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# NATIONAL CERTIFICATE (VOCATIONAL)

## WELDING
### NQF LEVEL 3

(6031033)

**21 November 2019 (X-Paper)**
**09:00–12:00**

**This question paper consists of 6 pages.**

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**TIME: 3 HOURS**
**MARKS: 100**

## INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Use a pen with only BLUE or BLACK ink.
6. Write neatly and legibly.

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## QUESTION 1

Explain the terminology used to describe welded joints:

1.1 Fillet joint (3)

1.2 Single V-butt weld (2)

1.3 Penetration (3)

1.4 Weld face (1)

1.5 Weld length (1)

**[10]**

## QUESTION 2

2.1 FIGURE 1 shows a welding symbol diagram. Label each welding symbol and write the answer next to the letter (A–J) in the ANSWER BOOK.

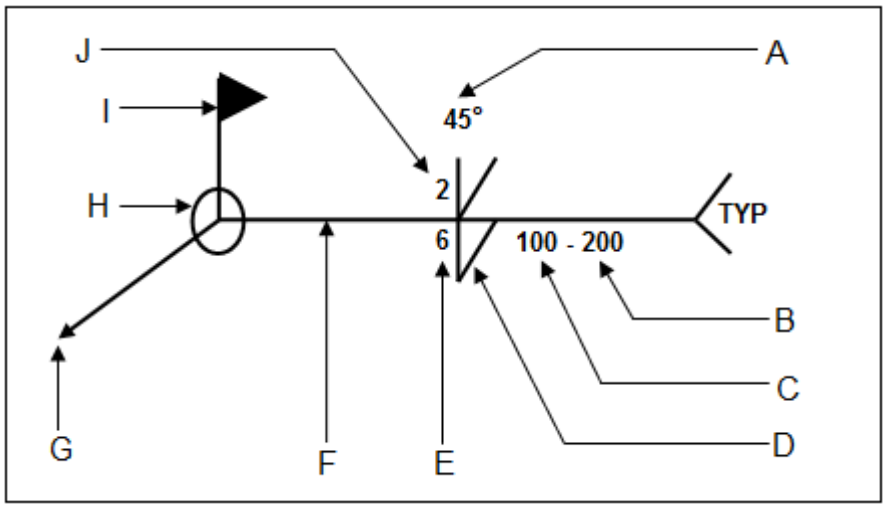



**FIGURE 1** (10 × 1) (10)

2.2 List FOUR causes of weld splatter during the shielded metal arc welding process (SMAW). (4)

2.3 List SIX visual inspections that have to be done after shielded metal arc welding (SMAW) has been completed. (6)

2.4 Name TWO methods of care that must be exercised on welding equipment after shielded metal arc welding (SMAW) has been completed. (2)

2.5 State THREE reasons for reporting faulty tools that are used in shielded metal arc welding (SMAW). (3)

**[25]**

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## QUESTION 3

3.1 Explain how you would eliminate the following hazards regarding oxy-acetylene gases:

3.1.1 Gas cylinder breach (2)

3.1.2 Flashback (4)

3.1.3 Oxygen explosion (2)

3.1.4 General fire hazard (2)

3.2 FIGURE 2 shows different types of oxy-acetylene flames. Label each type of flame next to the (A–C) in the ANSWER BOOK.

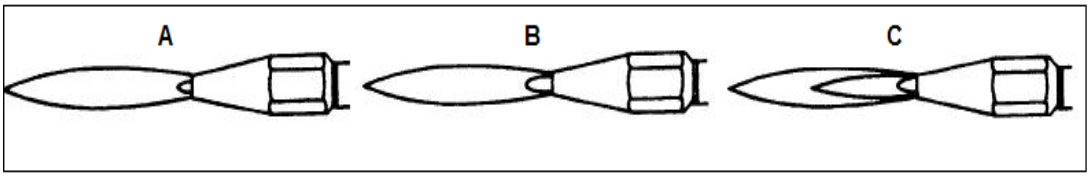



**FIGURE 2** (3)

3.3 Give SEVEN steps that must be followed when shutting down oxy-acetylene welding equipment. (7)

**[20]**

## QUESTION 4

4.1 Indicate whether the following statements are TRUE or FALSE with regard to gas tungsten arc welding (GTAW). Choose the answer and write only 'True' or 'False' next to the question number (4.1.1–4.1.10) in the ANSWER BOOK.

4.1.1 We refer to polarity when the electrons flow from the positive pole to the negative pole.

4.1.2 The weld bead is the result of the filler metal being deposited from a single pass.

4.1.3 Weld metal is the meeting point of two metals that are joined together.

4.1.4 Direct current is the flow of an electric charge that moves in one direction.

4.1.5 The arc length is the distance from one end of the weld to the other end.

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4.1.6 The weld pool is the area of the weld at which the base metal has reached its melting point and is ready to be mixed with the filler material.

4.1.7 Arc voltage is the current that flows through the circuit during the welding process.

4.1.8 An electrode is an electrical conductor used to conduct a current through a work piece in order to join two pieces together.

4.1.9 Alternating current is the amount of current present between the electrode and the work piece.

4.1.10 Shielded gas is an inert gas used during welding to protect the weld area from atmospheric gases.

(10 × 1) (10)

4.2 List FIVE safety concerns that must be considered when welding aluminium using gas tungsten arc welding (GTAW). (5)

4.3 State TWO safety precautions that must be considered when storing the following gas tungsten arc welding (GTAW) equipment:

4.3.1 Shielding gas cylinders

4.3.2 Electrodes

4.3.3 Filler rods

4.3.4 Gas nozzles and torches

4.3.5 Hose assembly

(5 × 2) (10)
**[25]**

## QUESTION 5

5.1 FIGURE 3 shows a metal inert gas (MIG) welding set-up diagram. Label each letter (A–J) in the ANSWER BOOK.

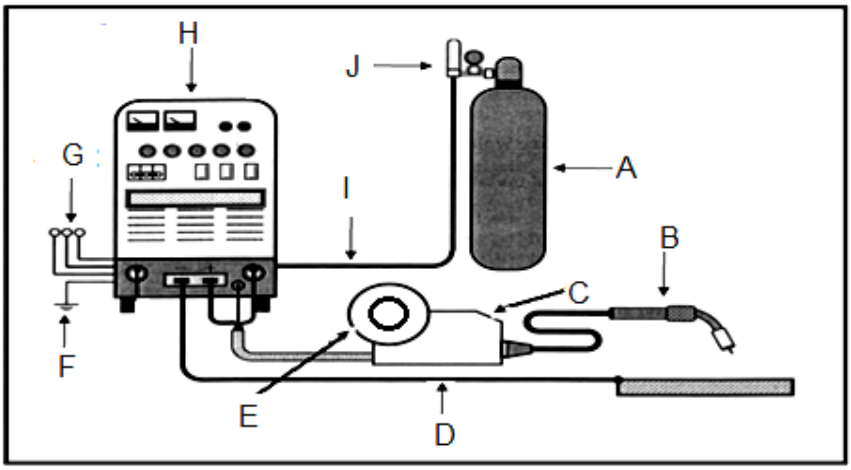



**FIGURE 3** (10 × 1) (10)

5.2 Calculate the amperage required to weld a 12-mm plate.

HINT: 0,0254 mm = 1 A (5)

5.3 List THREE types of tools required to check for alignment on a work piece. (3)

5.4 State TWO welding techniques that can be applied when performing MIG welding. (2)

**[20]**

**TOTAL: 100**

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